

Module 03: Assignment 01 - Project 01 - Franchise Value Question

Nischal Pant

4 billion dollar investment.

Analyzing our dataset:

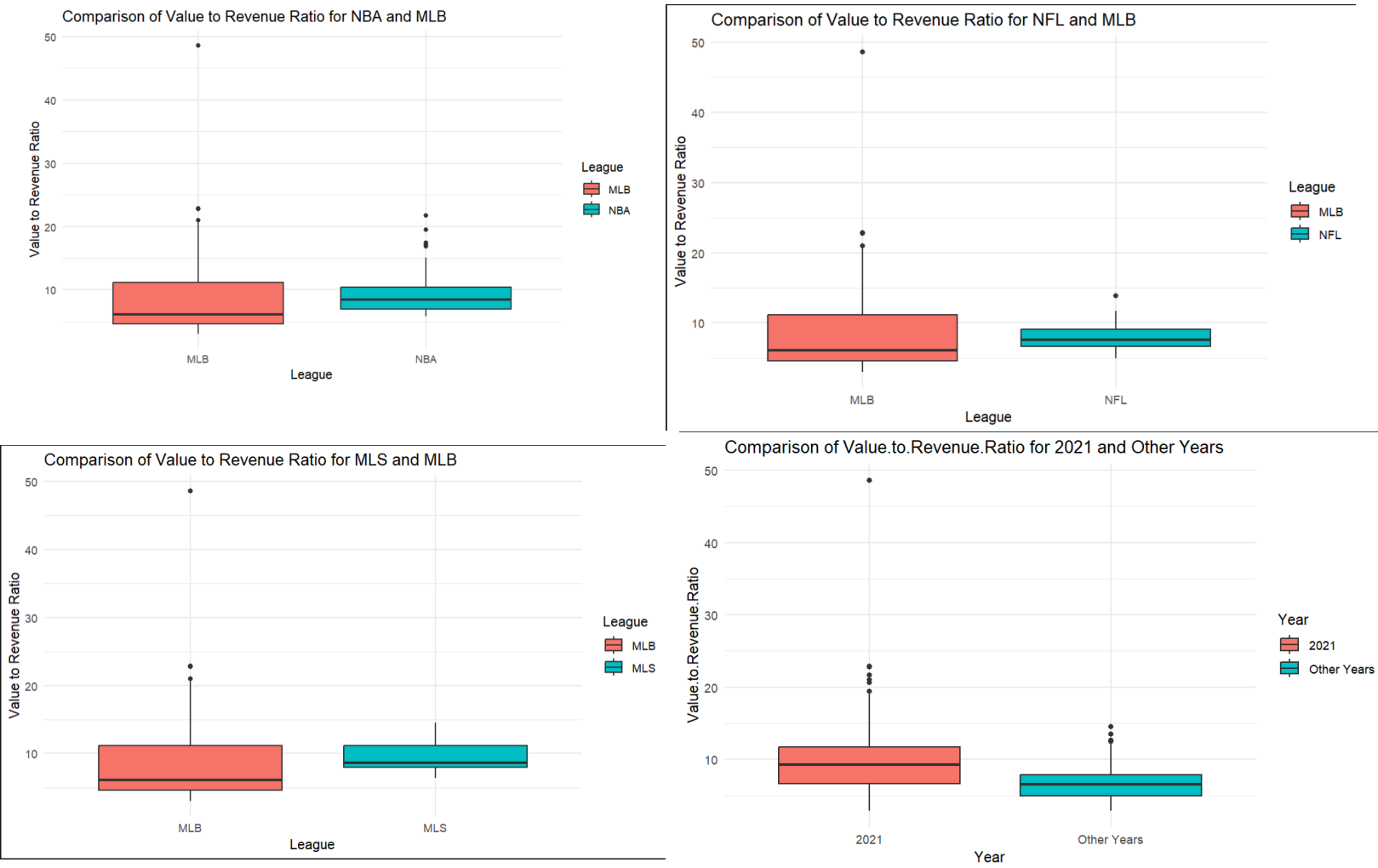
Variables:

- [1] "Year"
- [2] "League"
- [3] "Team"
- [4] "Revenue"
- [5] "Value"
- [6] "Year.over.Year.Value.Increase"
- [7] "Debt.Value"
- [8] "Income"
- [9] "Population"

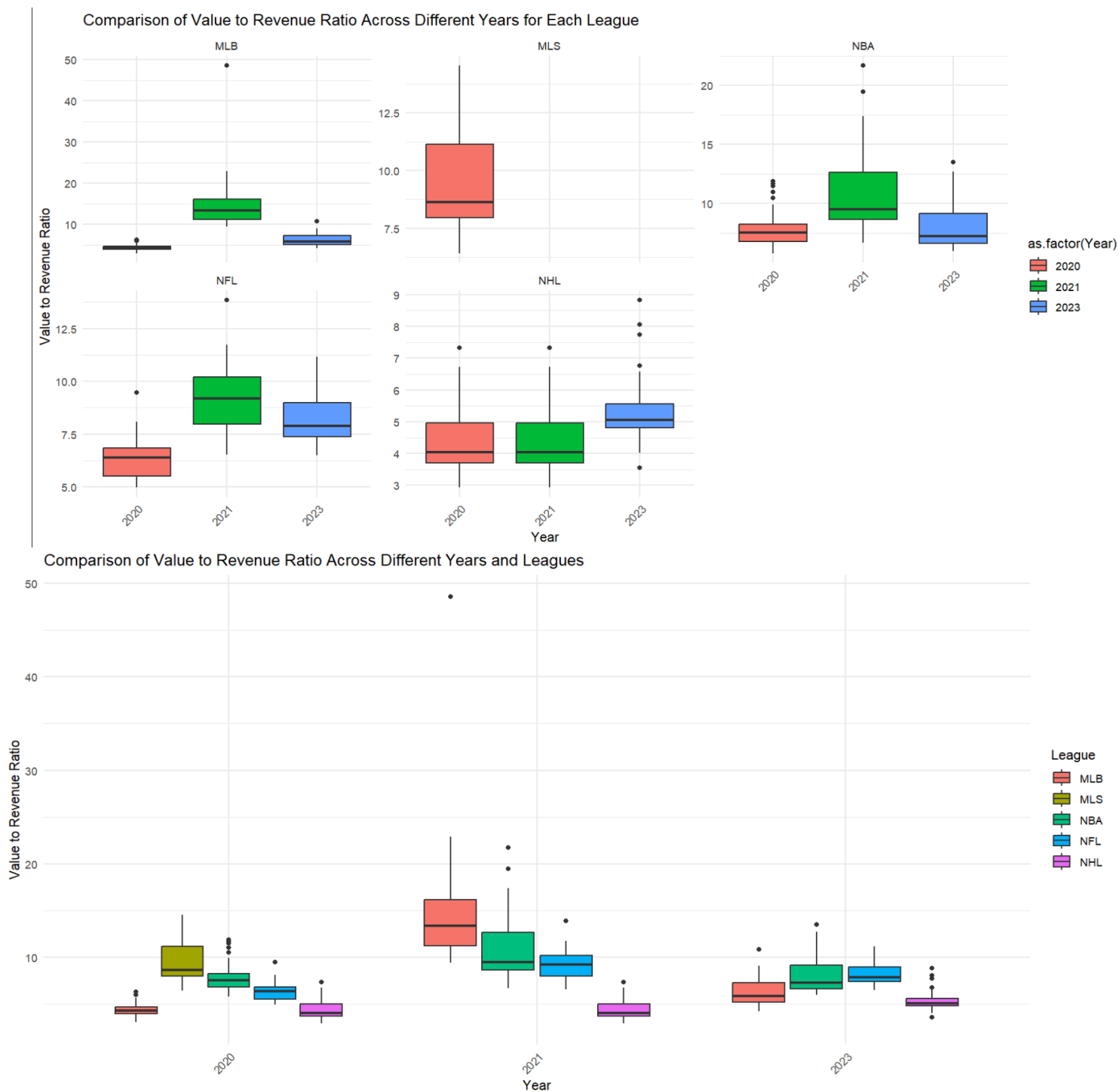
First, we have to clean our dataset—the variable Debt.Value has percentage value and decimal points so firstly we must convert them all into decimal points. If a value has a % sign we divide it by 100.

The income and revenue variables have input values as characters with “,” so we have to remove them specifically from the cells only because the file is comma-separated.

Compare the value-to-revenue ratios of MLB to the other sports leagues



We can first do some data exploration to look at the value to revenue ration for all the teams across the dataset. First by creating box plots shown above, we can see that all the league have a higher median value to revenue ratio than MLB teams.



Looking at other box plots, most of these comparisons are quite inconclusive as they are different all across. Further analysis and modeling is required to form decisions.

```

Call:
lm(formula = formula, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.185e+09 -4.567e+08 -1.596e+08  2.942e+08  3.645e+09

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.527e+11  1.252e+11   5.211 4.05e-07 ***
Year        -3.230e+08  6.196e+07  -5.213 4.02e-07 ***
Revenue      1.055e+01  7.228e-01  14.596 < 2e-16 ***
Income      -2.899e+00  1.175e+00  -2.466  0.0144 *
LeagueNBA    3.588e+08  1.510e+08   2.376  0.0183 *
LeagueNFL   -2.394e+08  1.816e+08  -1.319  0.1886
LeagueNHL   -3.778e+08  1.615e+08  -2.339  0.0202 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 707800000 on 239 degrees of freedom
(147 observations deleted due to missingness)
Multiple R-squared:  0.799,    Adjusted R-squared:  0.794
F-statistic: 158.4 on 6 and 239 DF,  p-value: < 2.2e-16

            GVIF Df GVIF^(1/(2*Df))
Year      1.885192  1      1.373023
Revenue  5.924457  1      2.434021
Income   3.653561  1      1.911429
League   4.690983  3      1.293830

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```

Lets do a linear regression for our dataset to get some stats:

Statistical Significance:

The 'Year,' 'Revenue,' 'Income,' and 'LeagueNHL' variables are statistically significant

Multiple R-squared: 0.799 (proportion of variance in 'Value' explained by the model).

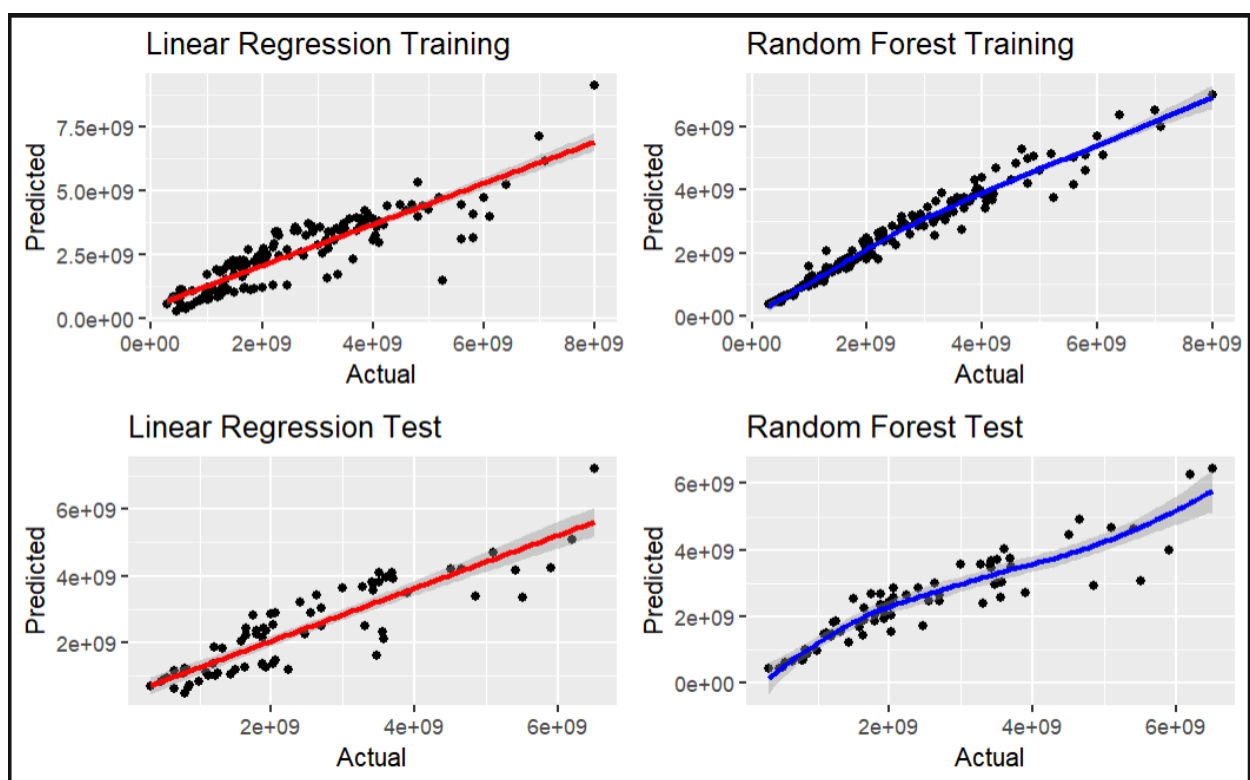
Adjusted R-squared: 0.794 (accounts for the number of predictors in the model).

F-statistic: 158.4 with a very low p-value ($< 2.2e-16$), suggesting that at least one predictor variable is significantly related to the response variable.

VIF values are very acceptable and indicate low multicollinearity (correlation between predictor variables). VIF values for 'Year,' 'Revenue,' 'Income,' and 'League' are provided.

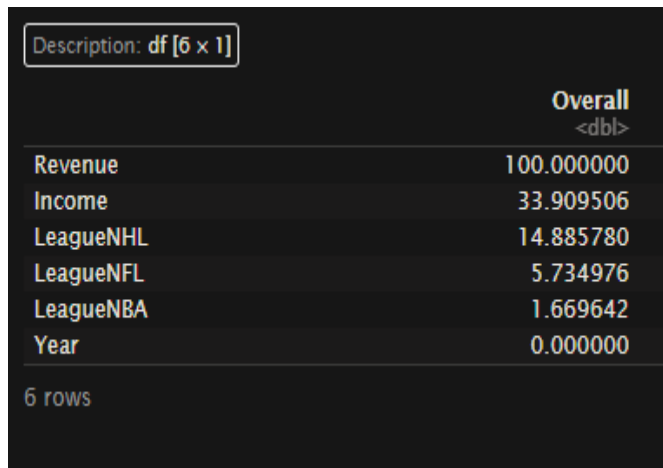
Conclusion:

The model suggests that '**Revenue**' has a positive impact on 'Value,' while 'Year' and 'Income' have a negative impact. The 'League' variable also contributes to predicting 'Value.' The model has a reasonable fit, explaining about 79.9% of the variance in 'Value.' There are many missing values so much further investigation is required. (147 deleted).



To further understand our dataset and hopefully make data driven decisions through modeling we have to first split the data into a training and test set. Then we can train a linear regression model asweel as a

random first model. We then plug these values into a table and visualize the results which can be observed above.



	Overall <dbl>
Revenue	100.000000
Income	33.909506
LeagueNHL	14.885780
LeagueNFL	5.734976
LeagueNBA	1.669642
Year	0.000000

6 rows

The output above shows the variable importance for the random forest model. In our case these independent variables are shown in percentage by their importance in predicting the accuracy of the model. Higher the percentage, higher the importance.

Revenue is considered the most important predictor in the random forest model. It means that changes in the Revenue variable have the most significant impact on the model's predictions of value. It is at 100. Then we can see that income is next with 33.9% which is still considered significant. There are other variables as well but for choosing teams for our investment portfolio we will not be using them.

We have conditions we must meet when we invest our 4 billion dollars. We can invest in multiple teams but only one from each league.

Our linear regression from the initial shows that revenue has the positive correlation and upon further modeling we can see revenue is in fact significant in predicting value.

Revenue is the most important predictor so we have to go with the team with the highest revenue the Dallas Cowboys of 2023 and 2021 from the NFL and 2023 Golden State Warriors in the NBA. If we would consider a team based on income we would still pick the Dallas Cowboys.

Considering how big the Dallas Cowboys have been and the amount of success they have generated over the years, investing all of our 4 billion dollar is not a bad idea. In order to ensure all our investment does not go down in one ship we should diversify. **1 billion shall be invested in GS warriors and 3 billion in Dallas cowboys.**